

Kai-Chi Weng

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EDUCATION

University of Michigan

Ann Arbor, MI

BSE in Robotics Engineering, Minor in Computer Science

Aug 2024 – Expected May 2027

GPA: 4.00/4.00

Related Coursework: Introduction to Machine Learning (EECS 445), Foundations of Computer Vision (EECS 504)

TECHNICAL & RESEARCH EXPERIENCE

Distributed Autonomous Systems and Control (DASC) Lab University of Michigan

Ann Arbor, MI

Undergraduate Research Assistant

Mar 2026 – Present

- Conduct literature reviews and theoretical studies in robot dynamics, advanced control theory, and safety-critical planning to build foundational knowledge for autonomous systems research
- Extend Dynamic Parabolic Control Barrier Functions (DPCBF) formulations from 2D to a 3D paraboloid constraint, implementing and benchmarking QP-CBF and MPC-CBF frameworks against existing techniques

NiFT team | Perot Jain TechLab at Mcity University of Michigan

Ann Arbor, MI

Software Lead

Jan 2026 – Expected Dec 2026

- Architected a 6-node ROS 2 autonomous driving workspace and a React-based live telemetry dashboard, enabling remote teleoperation and the transition of core node logic between Gazebo simulations and physical hardware
- Integrated a dual-antenna Mcity RTK GNSS with a custom 3D LiDAR pipeline, leveraging Open3D RANSAC and DBSCAN clustering to segment ground data and broadcast dynamic obstacles
- Engineered a decoupled control stack, piping a 10 Hz Pure Pursuit and CBF safety filter into a 20 Hz ADRC loop to map control inputs directly to 5-byte CAN messages

SIM-26 team | Multidisciplinary Design Program University of Michigan

Ann Arbor, MI

Subteam Lead

Jan 2026 – Expected Dec 2026

- Lead a 4-person subteam to design a "Wizard of Oz" driving simulation, engineering a multiplayer server in Unreal Engine 5 (C++/Blueprints) while delegating tasks and presenting weekly progress to a 20-person research group
- Formulate a literature-backed experimental methodology to eliminate human-subject social bias, utilizing Sample Entropy to quantify how lead vehicle unpredictability impacts driver visual demand

Michigan Mars Rover Team University of Michigan

Ann Arbor, MI

Autonomy & Robotic Arm Subteam Member

Sep 2024 – Dec 2025

- Implemented REP-105 sensor fusion via ROS 2, integrating multi-modal suites (RTK, IMU, CV) to linearize geodetic GPS data for centimeter-level autonomous navigation
- Developed a 6-DoF pose estimation pipeline using OpenCV and ArUco markers for precise fiducial tracking, reducing positional drift by 75%

WORK EXPERIENCE

Robotics Department University of Michigan

Ann Arbor, MI

Instructional Aid | ROB 320, ROB 201

May 2025 – Present

- Instruct students in applied calculus and robotics software engineering, teaching Julia for differential equations alongside C/C++ for Linux IPC, RIX middleware, coordinate transforms, and applied kinematics (URDF, FK/IK)
- Design comprehensive course exams, lead hands-on lab sections, and hold weekly office hours supporting 60+ students in debugging complex lower-level software architectures

Math Learning Center University of Michigan

Ann Arbor, MI

Math 115/116/215 Tutor

Jan 2025 – Jan 2026

- Deliver 10 hrs/week of drop-in and group tutoring for Calculus I-III
- Facilitate weekly Math 215 study groups with GSIs, supporting 20–30 students with structured problem sets

SKILLS

Programming Languages: C/C++, Python, MATLAB, R, Julia, Verilog, Assembly

Libraries & Frameworks: ROS 2, PyTorch, OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn

Software & Tools: Unreal Engine 5, Gazebo, Siemens NX, ModelSim, Linux, Git, Docker